

Introduction to Sufficient Statistics

Pascal Michailat
<https://pascalmichailat.org/w/>

What is the sufficient - statistic approach?

- Macro: policy analysis is structural $\rightarrow$ assume the entire structure of the economy & study policy within this structure.
- market structure is assumed almost religiously, without looking at evidence, but determines all policy results $\rightarrow$ policy insights are baked into initial assumptions about market structure
- assume Walrasian markets $\rightarrow$ markets are efficient, policy is unnecessary
- assume monopolistic markets $\rightarrow$ markups $> 0 \rightarrow$ price too high, not enough activity
- assume monopsonistic markets $\rightarrow$ markdowns $> 0 \rightarrow$ price too low, not enough activity
- Macro: policy analysis is structural $\rightarrow$ need to calibrate all model parameters (to run simulations)
- But many parameters are unobservable, cannot be properly identified & measured / estimated

• To address these issues $\rightarrow$ take sufficient-statistic approach.

- from public finance (1950s, 1960s, 1970s)

- Saez (2001) : optimal taxation

Chetty (2006) : $U \pm$

Chetty (2009) : survey

- new in mans in 2010 $\rightarrow$ spreading rapidly now.

Overview of sufficient-statistic approach:

① Minimal set of assumptions on welfare

function + structure of economy that allow for policy analysis $\rightarrow$ no need to specify entire model.

- analysis applies to a range of model (any model satisfying minimal set of assumptions)
- more likely to hold in real world & be useful to practitioners

② Express results & optimal policies in terms of statistics that can be properly estimated $\rightarrow$ connection b/w theory & empirics